SCALA: Asynchronous Acoustic Localization via Closed-Form AOA–TDOA Fusion

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***Abstract*—Acoustic indoor positioning systems (AIPS) have attracted considerable attention due to their low deployment cost and high localization accuracy. However, existing systems often rely on precise synchronization, suffer from limited robustness and update rates in complex environments, and lack efficient solutions for joint position and orientation estimation. In this paper, we propose SCALA, a temporal Self-Calibrated Asynchronous Localization Algorithm for passive acoustic pose estimation for mobile platforms. SCALA first introduces a batch signal processing pipeline that improves measurement robustness while supporting efficient multi-anchor processing. It then performs online inter-anchor self-calibration directly from AOA-derived geometric observations, enabling consistent TDOA construction without explicit synchronization. Based on the calibrated measurements, we further develop a closed-form hybrid AOA-TDOA solver that jointly estimates target position and orientation without iterative optimization, achieving both high efficiency and stable orientation observability. Experimental results in real-world indoor environments demonstrate that SCALA achieves a median localization error of $0.0215\,\mathrm{m}$ and a median orientation estimation error of 0.27°, validating its effectiveness for low-latency mobile acoustic pose estimation in practical indoor environments.**

***Index Terms*—Acoustic Positioning system, Asynchronous systems, Angle of Arrival, Time Difference of Arrival.**

I. Introduction

DRIVEN by the growing demand for applications such as emergency rescue [1], intelligent indoor navigation [2], mobile robot navigation and perception in degraded environments [3] and VR/AR services [4], indoor positioning and navigation systems have become an active research topic in recent years [5]–[8]. Existing indoor positioning system employ various sensing modalities, including vision [9], Wi-Fi [6], [7], BLE [10], UWB [11], RFID [12], visible light [13], and acoustic signals [14], [15]. Among them, AIPS are particularly attractive due to their low cost, high accuracy, and compatibility with existing audio infrastructure [8], [16], [17].

Current AIPS mainly rely on TOA/TDOA measurements [18], where distances are estimated from signal propagation time and positions are recovered through geometric optimization. Although these methods achieve high localization accuracy with relatively low computational complexity, they generally require strict clock synchronization across devices [18], [19], which limits practical deployment.

With the development of array signal processing, AOA-based localization methods have also gained increasing attention [5], [20]–[22]. AOA-based systems naturally support asynchronous operation and are more compatible with commodity acoustic hardware. However, the positioning error induced by angular estimation uncertainty increases nonlinearly with the receiver-to-anchor distance [23]. Moreover, since AOA measurements are defined in the local array coordinate system, receiver orientation becomes intrinsically coupled with pose estimation, further complicating localization [21]. Such pose information is particularly important for mobile robotic platforms, where orientation directly affects navigation, trajectory control, and environment interaction.

Therefore, integrating AOA and TDOA provides a complementary strategy that combines the asynchronous robustness and orientation observability of AOA with the metric constraints of TDOA, enabling deployable and high-precision acoustic pose estimation systems [24]–[30]. This capability is particularly desirable for mobile platforms that require both accurate localization and reliable orientation awareness under low-latency constraints. Nevertheless, effectively utilizing AOA and TDOA observations to construct an accurate and robust AIPS in real-world indoor environments still poses several fundamental challenges:

Challenge 1: How can TDOA constraints be reliably constructed under asynchronous conditions without explicit synchronization or external calibration?

Although AOA-based localization naturally avoids strict synchronization requirements, TDOA constraints still require consistent temporal references across anchors. In practice, unknown inter-anchor transmission offsets prevent direct construction of valid TDOA equations. To address this problem, we propose an online self-calibration method driven by AOA observations. By exploiting AOA-based pose initialization and multi-frame aggregation, the method estimates effective inter-anchor timing offsets and compensates for asynchronous

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TABLE I
COMPARISON OF REPRESENTATIVE ACOUSTIC IPNS SYSTEMS WITH RESPECT TO THE KEY CHALLENGES.

Study	Measure Type	Sync-free	Server-free	Position Est.	Orientation Est.	Closed-form Solution	Accuracy
ASSIST [31]	TDOA	✗	✗	✓	✗	✗	<0.30 m
Guoguo [16]	TOA	✗	✗	✓	✗	✓	0.06 – 0.25 m
ARABIS [32]	TDOA	✓	✗	✓	✗	✗	<0.20 m
AALTS [18]	TDOA	✓	✗	✓	✗	✗	<0.49 m
SAILoc [33]	DoA + Dist. Est.	✓	✗	✓	✗	✓	<0.60 m
MOM [34]	AoA	✓	✓	✗	✓	✗	2° – 20°
RAILS [20]	AoA	✓	✗	✓	✗	✓	0.10 m
[35]	AoA + Wheel Odom.	✓	✓	✓	✗	✗	0.19 m
SCALA (Proposed)	AoA + TDOA	✓	✓	✓	✓	✓	0.0215 m / 0.27°

clock biases, enabling reliable TDOA construction under fully asynchronous conditions.

Challenge 2: How can AOA and TDOA measurements be fused into a unified formulation with a closed-form solution for low-latency pose estimation?

As shown in Tab. I, most existing AIPS rely on iterative numerical optimization, resulting in high computational cost. Existing closed-form methods also often neglect the coupling between orientation variables and AOA observations in the array coordinate system, limiting their applicability in practical AIPS. To overcome this limitation, we develop a weighted AOA-TDOA fusion formulation with an adjustable confidence factor that jointly models position and orientation variables. Based on this formulation, we derive a closed-form solution for efficient pose estimation without requiring additional pose sensors.

Challenge 3: How can multi-anchor AOA and TDOA measurements be extracted efficiently for low-latency asynchronous acoustic localization?

AOA-based localization methods typically rely on high-quality multi-channel audio signals, where channel-wise preprocessing introduces redundant computation. Although TDOA-based methods operate on single-channel observations, they still lack an efficient mechanism to jointly extract measurements from concurrent multi-anchor signals. To address this issue, we design an frequency division multiple access (FDMA) signal scheme for structured separation of multi-anchor transmissions. Based on this design, we propose an FMCW-based batch TOA estimator (FBTE) that extracts TDOA measurements for all anchors in a unified process, avoiding repeated per-anchor operations. In addition, we introduce a time aligned batch FrFT filtering module (TABF) that jointly processes multi-channel data for AOA estimation, significantly improving efficiency while maintaining robustness.

The main contributions of this work are summarized as follows:

- We propose SCALA, a temporal self-calibrated asynchronous localization framework that jointly integrates AOA and TDOA constraints for low-latency acoustic pose estimation.
- We propose an AOA-guided on-the-fly self-calibration mechanism that estimates inter-anchor timing offsets directly from geometric observations, enabling consistent TDOA construction without explicit synchronization.
- We formulate a unified hybrid AOA-TDOA model and derive a closed-form solution for efficient joint position and orientation estimation for low-latency mobile acoustic platform without iterative nonlinear optimization.
- We develop an FDMA-based signaling scheme together with two batch signal processing modules, FBTE and TABF, for efficient multi-anchor TOA and AOA extraction with reduced multi-band and multi-channel processing overhead.
- Real world experiments demonstrate that SCALA achieves a median localization error of 0.0215 m and a median orientation error of 0.27°, validating its accuracy and robustness.

II. RELATED WORKS

A. TOA/TDOA-Based AIPS

Due to their robustness and simple solving procedures, TOA- and TDOA-based methods are widely adopted in acoustic indoor positioning systems (AIPS) [16], [17], [36]–[39]. ALPS [38] employs wireless synchronization and TDMA signaling for TOA localization, while Guoguo [16] combines adaptive TOA estimation with non-line-of-sight identification to achieve 6–25 cm accuracy. In [17], GCC-based TDOA detection is integrated with EKF fusion for sub-meter localization, and [39] adopts a coarse-to-fine arrival time estimation strategy to achieve a 100th percentile error below 0.5 m.

Despite their promising accuracy, these methods typically process anchor signals independently and rely on additional synchronization modules, which introduce redundant computation and limit practical deployment on commercial off-the-shelf acoustic devices. To relax synchronization requirements as described in *Challenge 2*, several asynchronous TDOA systems have been proposed [18], [19], [32]. ARABIS [32] performs localization using periodically collected timestamps, achieving a 95th percentile error of 7.4 cm, while AALTS [18] employs orthogonal chirp modulation and server-side delay estimation to achieve a 90th percentile error of 0.49 m with four anchors. However, these methods still depend on additional observations, bidirectional communication, or centralized processing. To address this issue, we propose an AOA-guided on-the-fly anchor clock self-calibration mechanism for fully passive asynchronous localization.

B. AOA-Based AIPS

AOA-based localization systems have been extensively studied in wireless positioning domains [6], [40]–[42]. Due to their inherent ability to operate without strict synchronization, AOA-based methods have also attracted increasing attention in AIPS in recent years [20], [33]–[35], [43], [44]. For instance, [35] proposes a direction-based indoor localization method using microphone pairs and asynchronous acoustic beacons, where cross-correlation functions are employed as likelihoods to suppress false detections, achieving 0.19 m 2D accuracy. SAILoc [33] combines AOA and distance estimation through frequency-domain time-delay analysis, achieving a maximum 2D localization error of 0.6 m. RAILS [20] utilizes a distributed microphone array network with ultrasonic chirp beacons and achieves an average 3D localization accuracy of approximately 0.1 m.

Despite these advances, existing methods are often limited by array signal processing complexity or system design, and fail to exploit signal structure for unified multi-anchor batch processing, leading to redundant computation and failing to effectively address the efficiency issue in Challenge 1. To overcome this limitation, we design an FDMA signal scheme that enables structured separation of multi-anchor transmissions, and further develop FBTE to efficiently obtain TDOA measurements and TABF to jointly process multi-channel data for AOA estimation, thereby significantly improving measurement efficiency.

C. Hybrid Localization Methods

Since AOA-based localization relies on angular estimation, its positioning error increases as the distance between the target node and anchors grows. To alleviate this issue, many studies combine TDOA and AOA measurements to improve localization accuracy [25]–[30]. For example, [25] proposes a hybrid AOA-TDOA localization method for UAVs in non-line-of-sight environments. [26] formulates a non-convex weighted least squares problem for joint AOA-TDOA localization and velocity estimation. [27] presents a closed-form solution using multi-station TDOA and single-station AOA measurements and achieves the Cramér-Rao lower bound under small noise conditions. [28] adopts a modified polar representation to unify near-field and far-field AOA localization and extends it to hybrid AOA-TDOA localization. [29] proposes a theoretically optimal three-dimensional localization method using combined TDOA and AOA observations from only two anchors.

Despite their promising theoretical performance, these methods often lack validation in practical environments. More importantly, they neglect the coupling between the orientation variable and AOA observations in the array coordinate system, which limits their applicability to passive acoustic localization systems and prevents direct joint position-orientation estimation. To address this issue, we decouple the orientation variable from AOA observations and develop a unified AOA-TDOA fusion framework with a closed-form solution for low-latency joint pose estimation.

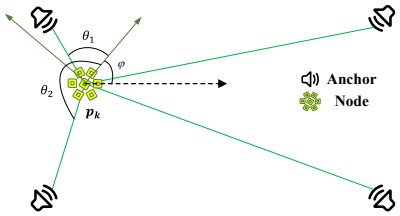


Fig. 1. SCALA measurement model

III. System Model

The proposed localization system, SCALA, consists of a target node equipped with a Uniform Circular Microphone Array (UCMA) and multiple spatially distributed asynchronous acoustic anchors. The target node passively receives periodic acoustic signals and performs batch processing to extract TOA and AOA measurements without requiring communication among devices. The corresponding measurement model is illustrated in Fig. 1.

SCALA enables asynchronous hybrid localization by combining online inter-anchor timing self-calibration with closed-form AOA-TDOA pose estimation, as illustrated in Fig. 3. Since 2D pose information is sufficient for most indoor scenarios, this work focuses on estimating the planar pose $p = (x, y, \varphi)$.

A. Transmitted Signals

LFM signals exhibit strong robustness to interference and multipath effects, which makes them widely adopted in AIPS. In addition, their linear frequency sweep property enables the use of FMCW techniques for efficient and accurate extraction of acoustic measurements. To exploit these properties while ensuring mutual interference avoidance among anchors and maintaining a high signal refresh rate, SCALA adopts an FDMA scheme to modulate LFM signals. To exploit these properties, SCALA adopts an FDMA scheme to modulate LFM signals. The signal transmitted by the i-th anchor is expressed as

$$s_i(t) = I \cdot \cos\left(2\pi\left(f_i t + \frac{B}{2T_c}t^2\right)\right), \quad 0 \le t \le T_c, \quad i = 1, 2, \cdots, M \tag{1}$$

where I denotes the signal amplitude, f_i is the initial frequency of the i-th anchor, B is the bandwidth, and T is the signal duration. In SCALA, the carrier frequencies satisfy $f_{i+1} = f_i + B$ to efficiently utilize the limited bandwidth of commercial acoustic devices. Fig. 2 illustrates the time-frequency representation of the transmitted signal under the proposed scheme. It is worth noting that this signal design represents one feasible configuration, and alternative designs can be adopted by adjusting the parameters without loss of generality.

B. Received Signal Model

To ensure lightweight deployment and ease of integration on mobile platforms, while maintaining unambiguous AOA

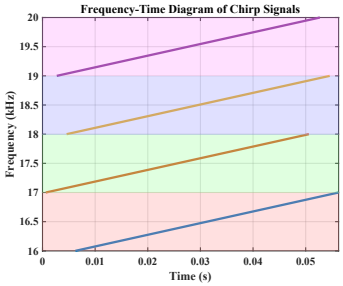


Fig. 2. Signal scheme adopted in the SCALA system. Since no synchronization mechanism exists among anchors, the transmitted signals from different anchors do not share a common starting time in the time domain.

estimation from all directions and approximately uniform gain over the angular domain, SCALA adopts a small-aperture circular microphone array as the acoustic receiver. Under this configuration, the received signal satisfies the far-field assumption. Therefore, the received signal model $Y(f_j)$ of the microphone array in the proposed system is expressed as

$$Y(f_j) = A(f_j)S(f_j) + N(f_j), \tag{2}$$

where $S_i(f_j)$ and $N(f_j)$ denote the frequency-domain representations of $s_i(t)$ and $n_n(t)$, respectively, and $A(f_j)$ denotes the array steering matrix at frequency f_j. By applying a direction-of-arrival estimation algorithm together with known array geometry, the azimuth incident angle $\hat{\theta}$ of the signals emitted from anchors at known locations can be obtained from the received signal $Y(f_j)$.

C. Problem formulation

We consider a two-dimensional acoustic localization scenario, where the target state is defined as $p = (x, y, \varphi)^{\mathsf{T}}$ representing the planar position and orientation of the target node. The system consists of M anchors with known locations that periodically transmit FDMA acoustic signals. Due to the absence of global synchronization, each anchor has an unknown transmission time offset $\{\tau_i\}_{i=1}^{M}$. The target node passively receives these signals and extracts AOA measurements $\{\hat{\theta}_{i,k}\}_{i=1}^{M}$ together with TOA measurements $\{\hat{t}_{i,k}\}_{i=1}^{M}$.

To compensate asynchronous timing offsets, SCALA first estimates effective inter-anchor clock biases and obtains calibrated TOA measurements

$$\tilde{t}_{i,k} = \hat{t}_{i,k} - \hat{\tau}_i, \tag{3}$$

where $\hat{\tau}_i$ denotes the estimated inter-anchor timing offset. The calibrated measurements are then used to construct consistent TDOA constraints. The final pose is estimated through joint AOA-TDOA optimization:

$$\min_{p} \sum_{k} \left(\lambda \mathcal{L}_{\mathrm{AOA}}(p) + (1-\lambda)\mathcal{L}_{\mathrm{TDOA}}(p)\right), \tag{4}$$

where $\lambda \in [0, 1]$ balances the contributions of angular and temporal observations.

IV. The Proposed Method

As illustrated in Fig. 3, SCALA consists of three stages. The first stage is the signal processing layer, including FBTE for TOA estimation and TABF for signal enhancement and AOA extraction. The second stage performs online inter-anchor clock offset calibration based on the extracted measurements. Finally, the pose estimation layer employs a hybrid AOA-TDOA solver to jointly estimate the target position and orientation.

A. FMCW-based Batch TOA Estimator (FBTE)

FBTE first constructs a long chirp signal $s_{\mathrm{ref}}(t)$ as a reference signal whose frequency sweep rate is identical to that of the transmitted anchor signals, while its starting frequency is aligned with the lowest sub-band, as illustrated in Fig. 4(a). The received signal $x(t)$ is then mixed with the reference signal through a dechirping operation:

$$y(t) = x(t)s^{*}_{\mathrm{ref}}(t). \tag{5}$$

According to the principle of frequency modulated continuous wave (FMCW) signals [45], each LFM sub-band emitted by an anchor is transformed after dechirping into a single-tone component. For the i-th sub-band with starting frequency f_i, the corresponding beat frequency satisfies

$$f_{\mathrm{beat},i} = (f_i - f_{\mathrm{ref}}) - \mu t_i, \tag{6}$$

where t_i denotes the TOA of the i-th anchor and μ denotes the chirp slope. Accordingly, TOA estimation is reformulated as a frequency estimation problem:

$$\hat{t}_i = \frac{f_i - f_{\mathrm{ref}} - f_{\mathrm{beat},i}}{\mu}. \tag{7}$$

This transformation exploits the physical structure of LFM signals and avoids estimating structurally similar arrivals independently for individual sub-bands, thereby improving both robustness and computational efficiency.

To estimate beat frequencies accurately and efficiently, FBTE employs the chirp z-transform (CZT), which provides high-resolution spectral analysis within a specified frequency range without increasing computational complexity. Given the dechirped signal $y(t)$, the magnitude spectrum $|Y(f)|$ is obtained through CZT, as illustrated in Fig. 4(b), and peak detection yields an initial beat frequency estimate $\hat{f}_{\mathrm{beat},i}$ for each sub-band. However, direct frame-wise peak selection is often unstable under multipath propagation due to competing spectral peaks. To improve temporal robustness, FBTE further introduces a temporally consistent estimation strategy with sub-bin refinement and cross-frame beat tracking, which suppresses abrupt peak switching across consecutive frames. To mitigate discretization errors introduced by finite spectral resolution, a quadratic interpolation is applied around the detected spectral peak. Let $|Y_{k-1}|$, $|Y_k|$, and $|Y_{k+1}|$ denote the magnitudes of three adjacent frequency bins around the detected peak. The sub-bin offset is estimated as

$$\rho = \frac{1}{2}\frac{|Y_{k-1}| - |Y_{k+1}|}{|Y_{k-1}| - 2|Y_k| + |Y_{k+1}|}, \tag{8}$$

and the refined beat frequency is obtained as

$$f_{\mathrm{beat},i} = f_k + \rho\Delta f, \tag{9}$$

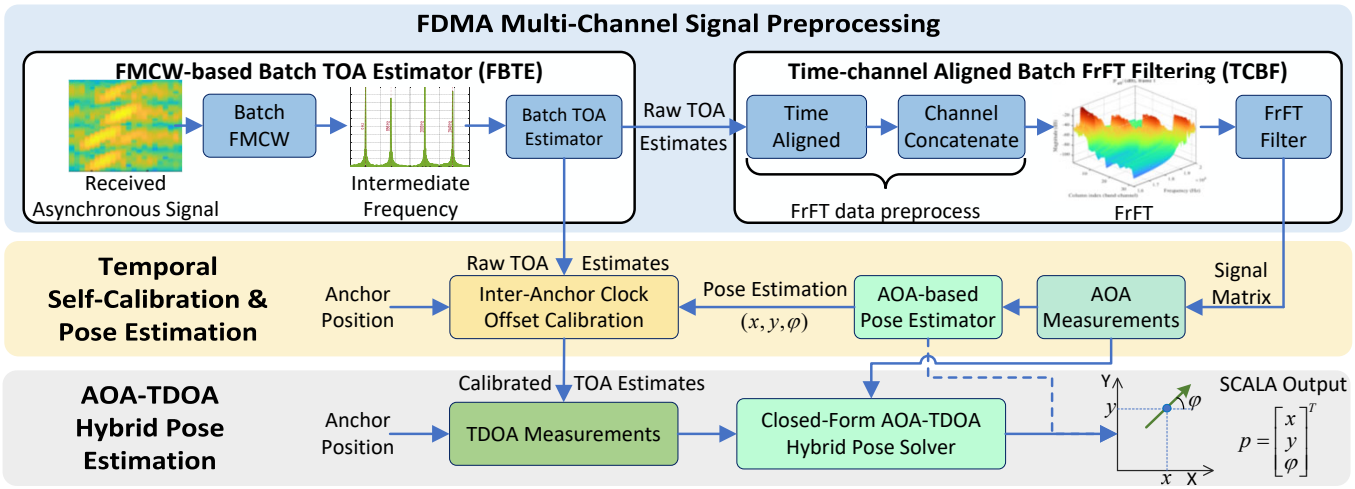


Fig. 3. Overview of the SCALA system. The target node equipped with an N-channel microphone array passively receives FDMA acoustic signals transmitted from M asynchronous anchors. FBTE first performs batch TOA estimation, and the estimated TOA values are further used for signal alignment and TABF-based AOA extraction. The estimated AOA and TOA measurements are then used for online inter-anchor offset calibration and consistent TDOA construction. Finally, a closed-form hybrid AOA-TDOA solver jointly estimates the position and orientation of the target node.

where Δf denotes the frequency resolution. This refinement reduces quantization-induced jitter and improves the smoothness of TOA estimates.

To further suppress peak switching caused by multipath interference, FBTE incorporates a cross-frame beat tracking strategy. The beat-frequency state is recursively updated as

$$f_{\text{state}}[j] = \gamma f_{\text{state}}[j-1] + (1-\gamma)\hat{f}_{\text{beat}}[j], \tag{10}$$

where $\hat{f}_{\text{beat}}[j]$ denotes the beat frequency estimated in the j-th frame and γ is a smoothing factor. For the subsequent frame, peak searching is constrained within a narrow frequency window centered at $f_{\text{state}}[j]$, allowing the estimator to favor spectrally consistent peaks while remaining adaptive to gradual variations.

This temporal constraint is well justified in our setting because in the indoor scenario the target node usually moves at a relatively low speed and the signaling interval between consecutive frames is short. As a result, the induced TOA variation between adjacent frames is limited, which ensures that the true beat frequency evolves smoothly and supports the validity of the tracking constraint.

By jointly exploiting sub-bin refinement and temporal beat tracking, the proposed scheme transforms independent frame-wise peak detection into a temporally constrained estimation process, effectively suppressing multipath-induced peak switching and improving the robustness of TOA estimation.

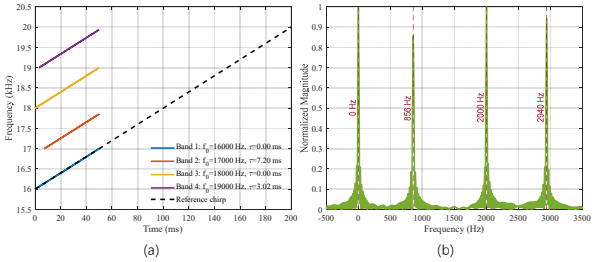


Fig. 4. An example of the intermediate process of the proposed FBTE. (a) Generated reference signal based on known parameters for an input signal containing four sub-bands. (b) CZT spectrum in the dechirped domain showing the beat frequencies of the four sub-bands.

Accordingly, FBTE serves as a front-end timing observable generator, producing a multi-anchor TOA vector

$$\hat{\boldsymbol{t}} = [t_1, t_2, \ldots, t_M]^\top, \tag{11}$$

which is used both for time-channel alignment in TABF and as temporal constraints for the subsequent asynchronous hybrid localization solver.

B. Time Aligned Batch FrFT Filtering (TABF)

According to the signal scheme in SCALA, M anchors transmit frequency-division multiplexed chirp signals over M sub-bands, while the receiving device employs a uniform circular microphone array with N elements. Based on the TOA estimates $\boldsymbol{t}$ obtained from the FBTE stage, a length-K signal segment associated with the i-th sub-band is synchronously extracted from the m-th microphone channel as

$$x_{i,m}[n] = s_m[n_i + n], \quad n = 0, \ldots, K-1 \tag{12}$$

where

$$n_i = \lfloor t_i f_s \rfloor, \tag{13}$$

and f_s denotes the sampling rate.

Instead of independently processing each anchor-channel pair, all aligned observations are organized into a structured time-channel batch matrix

$$\mathbf{X} = [\mathbf{x}_{1,1}, \cdots, \mathbf{x}_{1,N}, \cdots, \mathbf{x}_{M,N}] \in \mathbb{C}^{K \times MN}, \tag{14}$$

where each column corresponds to one aligned sub-band-channel observation. This formulation preserves temporal alignment across sub-bands and spatial consistency across channels, enabling coherent batch processing in the fractional domain. To exploit the energy concentration property of LFM signals, a shared-order FrFT is jointly applied to all columns of $\mathbf{X}$:

$$\mathbf{Y} = \mathcal{F}^{(\alpha)}\{\mathbf{X}\}, \tag{15}$$

where $\mathcal{F}^{(\alpha)}(\cdot)$ denotes the column-wise FrFT of order α. Since all transmitted chirps share the same sweep rate, all aligned

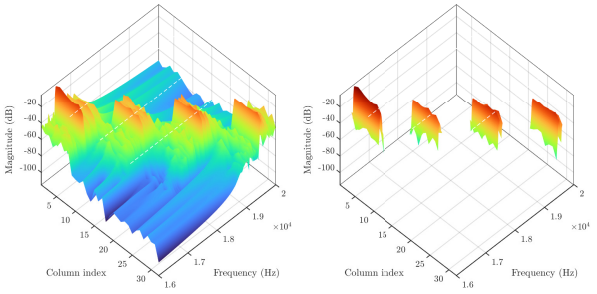


Fig. 5. Subfigures (a) and (b) illustrate the FrFT-domain representation of the received signal ($M = 4$, $N = 8$) before and after TABF filtering, respectively. Before filtering, indoor multipath propagation produces multiple secondary peaks in addition to the direct-path component, which degrades the reliability of acoustic measurements. After TABF filtering, most multipath components are effectively suppressed while the direct-path component remains dominant.

observations admit a common optimal fractional projection geometry. Therefore, the transform order is shared across all sub-bands and channels and is determined as

$$\alpha = \arctan\left(\frac{f_s^2 T}{B}\right), \tag{16}$$

where T and B denote the chirp duration and bandwidth, respectively.

By jointly processing all aligned observations using a shared transform order, TABF converts repeated per-anchor FrFT operations into a unified batch projection framework. This design preserves the coherent energy concentration of direct-path components while significantly reducing redundant preprocessing overhead in multi-anchor systems.

Under the optimal order α, the direct-path component is concentrated around a dominant fractional-domain peak. For the i-th sub-band, a peak-guided fractional-domain filter is defined as

$$H_{\alpha,i}(f) = \begin{cases} 1, & f_i - \frac{\Delta f}{2} \le f \le f_i + \frac{\Delta f}{2} \\ 0, & \text{otherwise}, \end{cases} \tag{17}$$

where f_i denotes the dominant peak frequency associated with the direct-path component. In practice, the peak with the maximum amplitude is selected as the direct-path peak. The filtered fractional-domain signal is then obtained as

$$\widetilde{Y}_i(f) = Y_i(f) H_{\alpha,i}(f). \tag{18}$$

By combining TOA-guided time alignment with shared-order batch FrFT projection, TABF establishes a structured fractional-domain filtering framework for robust AOA preprocessing. This formulation not only improves denoising robustness through coherent processing, but also substantially reduces the computational overhead of multi-channel signal preprocessing, making it well suited for low-latency real-time acoustic localization systems.

C. AOA-based Inter-anchor Clock Offsets Calibration

To address the asynchronous inter-anchor timing issue described in Challenge 1, we propose an AOA-based pose initialization and online self-calibration framework that jointly performs directional measurement acquisition, closed-form pose estimation, and timing-bias compensation through multi-frame aggregation without dedicated synchronization hardware.

AOA Measurement Acquisition. In SCALA, the MUSIC algorithm is independently applied to each sub-band, where the signal subspace dimension is set to one. The covariance matrix $\widetilde{\mathbf{R}}_i$ is constructed from the TABF-filtered signal $\widetilde{Y}_i(f)$ in the fractional Fourier domain. Through TABF processing, the broadband non-stationary LFM signals are transformed into stationary narrowband components with improved signal quality, thereby enabling robust AOA estimation using MUSIC. The MUSIC spectrum is evaluated over a predefined angular grid with step size $\Delta\theta$ to balance angular resolution and computational cost. To further reduce discretization errors caused by grid-based peak search, a lightweight Gauss–Newton refinement initialized from the spectral peak is applied to obtain sub-grid angular estimates. The direction of arrival associated with the i-th anchor is estimated as

$$\hat{\theta}_i = \arg\max_{\theta} \; P_{\mathrm{MUSIC}}\left(\theta; \widetilde{\mathbf{R}}_i\right), \tag{19}$$

where only the azimuth component is retained for the considered 2D pose estimation problem. Collecting all directional measurements yields the AOA observation vector

$$\hat{\boldsymbol{\theta}} = [\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_M]^\top, \tag{20}$$

which serves as the input to the subsequent closed-form pose initialization module.

To improve temporal consistency under multipath interference, an AOA peak tracking strategy is further introduced. Similar to the beat-frequency tracking adopted in the FMCW-TOA estimation stage, the directional estimate from the previous frame is used as a temporal prior to constrain the peak selection in the current frame. Given the MUSIC spectrum $P_t(\theta)$ at frame t, the AOA estimate is determined within a local angular neighborhood:

$$\begin{aligned} \hat{\theta}_t &= \arg\max_{\theta} \; P_t(\theta) \\ \text{s.t.} \quad & \hat{\theta}_{t-1} - \Delta\Theta \le \theta \le \hat{\theta}_{t-1} + \Delta\Theta. \end{aligned} \tag{21}$$

and $\Delta\Theta$ denotes the angular search half-width. This strategy improves the temporal continuity of directional measurements and suppresses occasional peak switching caused by multipath fluctuations while preserving real-time performance.

Closed-form AOA-based Pose Initialization. Since AOA observations are measured in the UCMA coordinate frame, they are jointly coupled with the target position and orientation, resulting in a nonlinear pose estimation problem. Existing methods typically rely on iterative optimization [21], [23], which often suffers from high computational cost and initialization sensitivity. To address this issue, we propose a closed-form AOA pose estimation method for efficient pose initialization in the subsequent asynchronous hybrid localization stage.

As illustrated in Fig. 1, the AOA measurements relate the target pose (x, y, φ) to the anchor locations (a_i, b_i). To obtain a closed-form solution, the resulting nonlinear equations are

reformulated into an equivalent linearized form by introducing the following auxiliary variables:

$$\begin{aligned} u_1 &= x\sin\varphi - y\cos\varphi, \\ u_2 &= x\cos\varphi + y\sin\varphi, \\ u_3 &= \sin\varphi, \\ u_4 &= \cos\varphi. \end{aligned} \tag{22}$$

Substituting these auxiliary variables into the measurement constraints yields a homogeneous linear system with respect to $\mathbf{u} = [u_1, u_2, u_3, u_4]^\mathsf{T}$. For the j-th anchor, the resulting constraint is

$$\begin{aligned} \cos\theta_i u_1 + \sin\theta_i u_2 - (a_j\cos\theta_i + b_j\sin\theta_i)u_3 \\ - (a_j\sin\theta_i - b_j\cos\theta_i)u_4 = 0. \end{aligned} \tag{23}$$

The detailed derivation is provided in the Appendix. Stacking all anchor constraints leads to the homogeneous linear system

$$\mathbf{M}_{\mathrm{AOA}}\mathbf{u} = \mathbf{0}, \tag{24}$$

where

$$\mathbf{M}_{\mathrm{AOA}} = \left[\mathbf{m}_1^\mathsf{T}, \mathbf{m}_2^\mathsf{T}, \ldots, \mathbf{m}_M^\mathsf{T}\right]^\mathsf{T}, \tag{25}$$

whose i-th row is

$$\mathbf{m}_i^\mathsf{T} = \left[c_i,\ s_i,\ -(a_i c_i + b_i s_i),\ -(a_i s_i - b_i c_i)\right], \tag{26}$$

with

$$c_i = \cos\theta_i, \qquad s_i = \sin\theta_i. \tag{27}$$

In the presence of measurement noise, the homogeneous constraint is only approximately satisfied. Therefore, we solve the problem using a constrained least-squares criterion:

$$\min_{\|\mathbf{u}\|_2=1} \|\mathbf{M}_{AOA}\mathbf{u}\|_2^2. \tag{28}$$

Since the homogeneous solution has scale ambiguity, the physical constraint

$$u_3^2 + u_4^2 = 1$$

is further imposed through normalization:

$$\mathbf{u} = \frac{\mathbf{u}_{\mathrm{raw}}}{\sqrt{u_{\mathrm{raw},3}^2 + u_{\mathrm{raw},4}^2}}. \tag{29}$$

The closed-form solution is then obtained from the right singular vector corresponding to the smallest singular value of $\mathbf{M}_{AOA}$, from which the pose estimate is recovered as

$$p^{AOA} = \begin{bmatrix} x \\ y \\ \varphi \end{bmatrix} = \begin{bmatrix} u_1 u_3 + u_2 u_4 \\ u_2 u_3 - u_1 u_4 \\ \mathrm{atan2}(u_3, u_4) \end{bmatrix} \tag{30}$$

This closed-form formulation avoids iterative nonlinear optimization while providing an efficient and reliable pose initialization for the subsequent asynchronous hybrid localization solver.

On the fly Inter-anchor Clock Offsets Calibration. We leverage the coarse pose estimate $p^{AOA} = [x, y, \varphi]^\mathsf{T}$ obtained from the closed-form AOA initialization together with the TOA observations estimated by FBTE to construct a geometric timing-bias model for asynchronous inter-anchor calibration. The estimated pose p^{AOA} provides geometric delay compensation, while the TOA measurements $\boldsymbol{t}$ provide temporal observations. Their discrepancy is used to estimate effective inter-anchor timing biases online without dedicated synchronization hardware.

The propagation delay from the target node to the i-th anchor is modeled as

$$\frac{r_i}{c} = \frac{d_i}{c} + \Delta_i + \epsilon_i, \tag{31}$$

where d_i denotes the geometric propagation distance inferred from the estimated pose, c is the sound speed, Δ_i denotes the unknown transmission timing offset of the i-th anchor, and ϵ_i represents measurement noise. Using the first anchor as reference, the relative timing observation satisfies

$$\Delta\tau_{i1} = \frac{d_i - d_1}{c} + \delta_{i1} + \epsilon_{i1}, \tag{32}$$

where $\delta_{i1} = \Delta_i - \Delta_1$ denotes the relative inter-anchor timing bias to be estimated.

Based on the geometric distance difference computed from $\mathbf{p}$, a single-frame timing-bias observation is constructed as

$$z_i^{(k)} = \Delta\tau_{i1}^{(k)} - \frac{\hat{d}_i^{(k)} - \hat{d}_1^{(k)}}{c} = \delta_{i1} + \varsigma_i^{(k)}, \tag{33}$$

where $\varsigma_i^{(k)}$ captures residual errors induced by pose uncertainty and measurement noise.

To improve robustness against frame-wise outliers, the timing-bias observations are statistically screened using a 2σ criterion. Given the empirical mean μ_i and standard deviation σ_i of $\{z_i^{(k)}\}_{k=1}^K$, only inlier observations satisfying

$$|z_i^{(k)} - \mu_i| \le 2\sigma_i \tag{34}$$

are retained to form the inlier set $\mathcal{I}_i$.

The effective timing bias is then estimated by solving a least-squares problem over the inlier set:

$$\hat{\delta}_{i1} = \arg\min_{\delta_{i1}} \sum_{k\in\mathcal{I}_i} \left(z_i^{(k)} - \delta_{i1}\right)^2, \tag{35}$$

whose closed-form solution is

$$\hat{\delta}_{i1} = \frac{1}{|\mathcal{I}_i|}\sum_{k\in\mathcal{I}_i} z_i^{(k)}. \tag{36}$$

This robust aggregation suppresses outlier frames caused by degraded AOA initialization or multipath-contaminated TOA estimates. Rather than recovering exact clock offsets, the proposed method estimates effective inter-anchor timing biases to mitigate dominant asynchronous errors in TDOA constraints. The AOA-based pose estimates are only used to calibrate the static transmission-time biases among anchors, thereby constructing pseudo-synchronized TDOA measurements for subsequent localization. Since the calibration is performed through multi-frame aggregation and statistical filtering, random AOA perturbations do not directly propagate into unstable TDOA constraints. In addition, the subsequent hybrid localization further regularizes residual calibration errors through the geometric consistency imposed by AOA observations.

D. Homogenized AOA-TDOA Hybrid Pose Estimator

Let the target pose be $\mathbf{p} = [x, y, \varphi]^\mathsf{T}$. To jointly linearize the coupled nonlinear constraints, we introduce the lifted variable

$$\begin{aligned} &u_1 = x\sin\varphi, u_2 = x\cos\varphi, u_3 = y\sin\varphi, u_4 = y\cos\varphi, \\ &u_5 = r_1\sin\varphi, u_6 = r_1\cos\varphi, u_7 = \sin\varphi, u_8 = \cos\varphi \end{aligned} \tag{37}$$

This homogeneous lifting embeds position, orientation, and TDOA nuisance variables into a unified representation.

For AOA observations, we reuse the same nonlinear bearing model introduced in the closed-form AOA initialization module, while reformulating it under the new lifted variables. Each AOA measurement yields a homogeneous linear constraint

$$\mathbf{m}_{i,\mathrm{AOA}}^\mathsf{T}\mathbf{u} = 0. \tag{38}$$

Stacking all AOA constraints gives

$$\mathbf{M}_{\mathrm{AOA}}\mathbf{u} = \mathbf{0}, \tag{39}$$

where

$$\mathbf{M}_{\mathrm{AOA}} = \left[\mathbf{m}_{1,\mathrm{AOA}}^\mathsf{T}, \ldots, \mathbf{m}_{M,\mathrm{AOA}}^\mathsf{T}\right]^\mathsf{T}, \tag{40}$$

whose i-th row is

$$\mathbf{m}_{i,\mathrm{AOA}}^\mathsf{T} = \left[c_i, s_i, s_i, -c_i, 0, 0, -(a_ic_i + b_is_i), -(a_is_i - b_ic_i)\right], \tag{41}$$

with

$$c_i = \cos\theta_i, \qquad s_i = \sin\theta_i. \tag{42}$$

For TDOA observations, let the differential range measurement relative to the reference anchor be

$$d_{i1} = r_i - r_1, \qquad i = 2, \ldots, M. \tag{43}$$

Following standard range-difference geometry, each TDOA constraint can be written as

$$2\Delta a_i x + 2\Delta b_i y + 2d_{i1}r_1 = H_i, \tag{44}$$

where

$$\Delta a_i = a_i - a_1, \quad \Delta b_i = b_i - b_1, \tag{45}$$

and

$$H_i = a_i^2 + b_i^2 - a_1^2 - b_1^2 - d_{i1}^2. \tag{46}$$

By projecting the above equation onto $\sin\varphi$ and $\cos\varphi$, each anchor pair generates two homogeneous linear constraints in the lifted space, which can be compactly written as

$$\mathbf{m}_{i,\mathrm{TDOA},\sin}^\mathsf{T}\mathbf{u} = 0, \qquad \mathbf{m}_{i,\mathrm{TDOA},\cos}^\mathsf{T}\mathbf{u} = 0. \tag{47}$$

Although Eq. (48) originate from the same TDOA relation, they are not redundant constraints. Instead, the two equations constrain complementary components of the lifted vector $\mathbf{u}$ through the $\sin\varphi$- and $\cos\varphi$-coupled subspaces, respectively. This construction enables the orientation-independent TDOA geometry to be consistently embedded into the same homogeneous lifted space as the AOA constraints while jointly constraining all lifted variables in the unified formulation. Stacking all projected constraints yields

$$\mathbf{M}_{\mathrm{TDOA}}\mathbf{u} = \mathbf{0}, \tag{48}$$

where

$$\mathbf{M}_{\mathrm{TDOA}} = \left[\mathbf{m}_{2,\sin}^\mathsf{T}, \mathbf{m}_{2,\cos}^\mathsf{T}, \ldots, \mathbf{m}_{M,\sin}^\mathsf{T}, \mathbf{m}_{M,\cos}^\mathsf{T}\right]^\mathsf{T}, \tag{49}$$

with

$$\mathbf{M}_{\mathrm{TDOA}} \in \mathbb{R}^{2(M-1)\times 8}.$$

The explicit coefficient forms of the projected TDOA rows are provided in the Appendix. The unified hybrid system is constructed as

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{\mathrm{AOA}} \\ \lambda\mathbf{M}_{\mathrm{TDOA}} \end{bmatrix}, \tag{50}$$

where λ serves as a confidence-balancing factor for heterogeneous AOA and TDOA constraints. In practice, we define a characteristic spatial scale as

$$L_{\mathrm{ref}} = \frac{1}{M}\sum_{i=1}^{M}\|\mathbf{s}_i\|_2, \qquad \mathbf{s}_i = [a_i, b_i]^\mathsf{T}, \tag{51}$$

and set

$$\lambda = \frac{\eta}{L_{\mathrm{ref}}}. \tag{52}$$

The parameter η controls the relative confidence between AOA and TDOA constraints. Increasing η assigns greater weight to TDOA measurements, while decreasing η places more emphasis on AOA constraints and improves robustness to TDOA bias. We observe that the performance of the proposed method remains stable over a wide range of η, and a sensitivity analysis of η is provided in the Appendix. The hybrid pose estimate is obtained by solving

$$\min_{\|\mathbf{u}\|_2=1}\|\mathbf{M}\mathbf{u}\|_2^2. \tag{53}$$

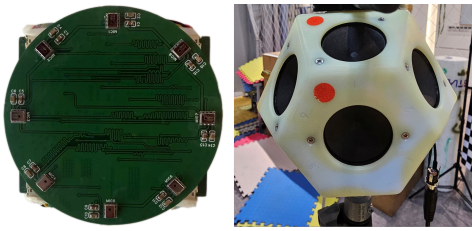
Due to measurement noise and model perturbations, the homogeneous system $\mathbf{M}\mathbf{u} = \mathbf{0}$ generally admits no exact non-trivial solution. Therefore, the localization problem is reformulated as the following constrained homogeneous least-squares optimization:

$$\min_{\|\mathbf{u}\|_2=1}\|\mathbf{M}\mathbf{u}\|_2^2 = \min_{\|\mathbf{u}\|_2=1}\mathbf{u}^\top\mathbf{M}^\top\mathbf{M}\mathbf{u}.$$

According to the Rayleigh–Ritz theorem, the optimal solution corresponds to the eigenvector associated with the smallest eigenvalue of $\mathbf{M}^\top\mathbf{M}$. Equivalently, under the singular value decomposition $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, the closed-form solution is obtained from the right singular vector corresponding to the smallest singular value of $\mathbf{M}$. The target pose can then be recovered from the estimated lifted vector as

$$\begin{aligned} \hat{x} &= u_1u_7 + u_2u_8, \\ \hat{y} &= u_3u_8 - u_4u_7, \\ \hat{\varphi} &= \mathrm{atan2}(u_7, u_8). \end{aligned} \tag{54}$$

Through homogeneous lifting and unified hybrid linearization, the proposed estimator transforms asynchronous AOA-TDOA localization into a fixed-dimensional eigenvalue problem, avoiding iterative nonlinear optimization while enabling efficient and robust joint estimation of position and orientation.



(a) (b)

Fig. 6. Acoustic devices used in the SCALA system: (a) an UCMA with 8 elements and a radius of 0.025 m; (b) a custom-designed speaker anchor for emitting acoustic localization signals. The anchor integrates multiple loudspeakers to allow flexible beam direction adjustment for different installation orientations and coverage requirements. In operation, only one loudspeaker is activated, while the others remain inactive.

V. Implementation

A. Experiment Setup

System Setup. We implement the complete SCALA system using commercially available acoustic hardware, as illustrated in Fig. 6. The system consists of $M = 4$ distributed anchors constructed using low-cost paper-cone loudspeakers and simple driver circuits, together with a receiver equipped with an $N = 8$ element uniform circular microphone array of radius $0.025\,\mathrm{m}$. This compact and cost-efficient configuration facilitates integration into mobile platforms while maintaining sufficient angular resolution within a limited aperture [46]. In addition, the circular geometry provides full azimuthal coverage and stable inter-element phase differences, improving the robustness of AOA estimation. The total hardware cost of the complete system is below 25 USD.

All devices operate at a sampling rate of $f_s = 48{,}000\,\mathrm{Hz}$. The localization signals are preloaded into each speaker and automatically played after power-up. The entire SCALA system is implemented on a computer equipped with an Intel i7-12700F CPU and 32 GB of RAM.

As described in Section III, the transmitted acoustic signals are designed to support reliable TOA estimation and effective multi-anchor separation while satisfying nonaudibility constraints and transducer characteristics. The corresponding system parameters are summarized in Table II.

Evaluation Scenario. We evaluate the proposed SCALA system in two representative indoor environments, as shown in Figs. 7a and 7b. Scene 1 is a relatively open area of size $8 \times 15\,\mathrm{m}^2$ with minimal obstacles, where four beacons are placed around the boundary at $(\mathbf{a}, \mathbf{b}) = \{(5.925, -2.293), (5.584, 3.548), (-5.940, 3.472), (-5.729, -2.219)\}$. All coordinates are defined in the motion capture coordinate frame. Scene 2 is a larger indoor hall of size $10 \times 17\,\mathrm{m}^2$ with reflective surfaces, including furniture and a glass curtain wall, resulting in strong multipath and reverberation. In this scenario, four beacons are deployed at $(\mathbf{a}, \mathbf{b}) = \{(0, 0), (6.6, 0), (6.6, 11.5), (0, 11.5)\}$, with coordinates defined in the SCALA system frame.

Ground truth in Scene 1 is obtained using an optical motion capture system with millimeter-level positional accuracy and sub-degree orientation accuracy, while Scene 2 uses LiDAR-based odometry via the LOAM algorithm [47], providing centimeter-level accuracy under experimental conditions. LiDAR is used solely for ground-truth acquisition, and the proposed system operates independently without requiring any camera or LiDAR sensors.

TABLE II
Parameters Configuration of the SCALA System

System Parameters	Value
I	1
f_s	48 kHz
f_1, f_2, f_3, f_4	16 kHz, 17 kHz, 18 kHz, 19 kHz
B	1 kHz
T_c	0.05 s
T_{guard}	0.05 s
K	1024
M	4
N	8
Δf	10
$\Delta\theta$	1°
$\Delta\Theta$	5°

B. Localization Baseline Methods.

AOA: We include the closed-form AOA-only method proposed in Section IV-C, which directly solves the AOA constraint equations in the array coordinate system to jointly estimate position and orientation. This baseline highlights the performance of pure AOA-based localization without TDOA constraints.

TDOA (GT-Calibrated): We further construct a TDOA baseline with externally calibrated clock offsets. A short trajectory segment is used to estimate inter-anchor offsets using ground-truth positions, and the calibrated offsets are then fixed for subsequent TDOA-based least-squares localization [17]. This baseline reflects the performance of synchronized TDOA localization.

AOA+TDOA (NLS): We also compare with a conventional hybrid nonlinear least-squares (NLS) method that iteratively solves the optimization problem defined by Eq. 4 to jointly estimate the target position and orientation. This method serves

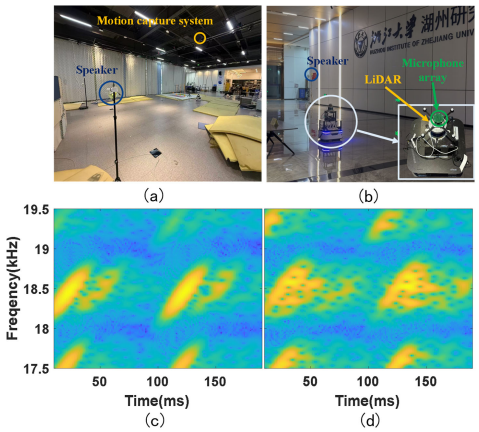


Fig. 7. (a) and (b) show the experimental setups of Scene 1 and Scene 2, respectively, while (c) and (d) present the corresponding received signals. Compared with Scene 1, Scene 2 exhibits stronger multipath interference and signal distortion. A guard interval $T_{guard} = 0.05\,\mathrm{s}$ is inserted between consecutive signal frames to reduce reverberation effects.

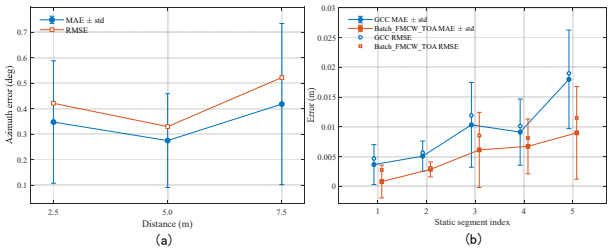


Fig. 8. Acoustic measurement accuracy results: (a) AOA estimation accuracy as a function of source-to-array distance. (b) Radial displacement estimation accuracy based on the FBTE module at different distances.

as a strong baseline for evaluating the effectiveness of the proposed closed-form formulation.

To ensure a fair comparison, all methods share the same signal processing pipeline and are evaluated under the same experimental conditions and performance metrics.

VI. Results

A. TOA Estimation Accuracy

Accurate TOA estimation is essential for reliable signal extraction, precise TDOA constraints, and overall localization performance in SCALA. To evaluate the accuracy of the FMCW-based constraints under practical conditions, we conduct controlled experiments in Scene 1, where the speaker remains fixed and the microphone array moves linearly from 0 to $2.6\,\mathrm{m}$. The estimated TOA values are converted into radial distances and compared with motion-capture ground truth.

As shown in Fig. 8(b), the proposed FBTE module consistently outperforms the GCC-based baseline in both accuracy and stability. The MAE remains within 0.008–$0.09\,\mathrm{cm}$ and the RMSE stays below $1.1\,\mathrm{cm}$, with tighter error distributions. This gain is attributed to the stabilized beat-frequency estimation and temporal tracking in the batch FMCW framework, which effectively reduces peak ambiguity and temporal jitter.

B. AOA Estimation Accuracy

AOA estimation accuracy directly affects the pose estimation performance of SCALA. To evaluate its effectiveness, experiments are conducted in Scene 1, where a speaker fixed at a height of approximately $2\,\mathrm{m}$ transmits signals to a rotating microphone array. The source-to-array distances are set to $2.5\,\mathrm{m}$, $5\,\mathrm{m}$, and $7.5\,\mathrm{m}$ to evaluate AOA accuracy under different propagation conditions, and the estimated AOA values are compared with motion-capture ground truth.

As shown in Fig. 8(a), the AOA estimation module of the SCALA system maintains high angular accuracy across different propagation distances. The MAE stays below $0.5°$ with relatively small variance, indicating stable and robust directional estimation. Specifically, at source-to-array distances of $2.5\,\mathrm{m}$, $5\,\mathrm{m}$, and $7.5\,\mathrm{m}$, the MAE values are $0.35°$, $0.28°$, and $0.42°$, respectively. Although AOA estimation accuracy is theoretically affected by signal attenuation at longer distances, the measured errors do not exhibit strictly monotonic behavior due to variations in multipath propagation and environmental noise under different spatial configurations. Nevertheless, the proposed method consistently maintains reliable AOA estimation performance.

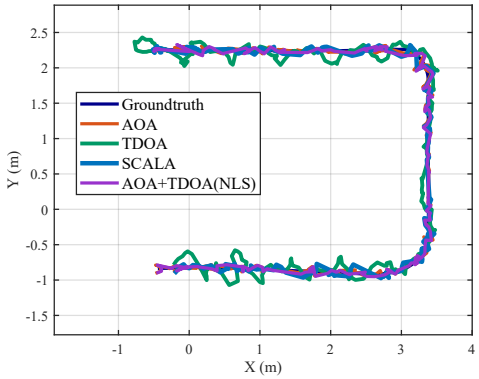


Fig. 9. The estimated and ground-truth trajectories of the node in Scene 1. To enhance visual clarity, the positions of the speakers are not shown in the figure.

C. Pose Estimation Accuracy

We evaluate the localization and orientation estimation performance of SCALA in Scene 1, which represents a typical indoor positioning scenario. As shown in Fig. 9, the TDOA-only baseline still exhibits noticeable trajectory fluctuations, due to multipath-induced TOA estimation errors. By incorporating AOA observations, both the AOA baseline and SCALA effectively suppress such localization jitter, since AOA measurements are less sensitive to temporal perturbations.

However, pure AOA-based methods suffer from nonlinear localization error growth as the distance between the target node and anchors increases. By introducing complementary TDOA constraints, both SCALA and the AOA-TDOA nonlinear least-squares (NLS) method achieve significantly improved localization accuracy. As reported in Fig. 10(a), SCALA achieves an MAE of $0.0293\,\mathrm{m}$, an RMSE of $0.0383\,\mathrm{m}$, a median error of $0.0215\,\mathrm{m}$, and a 90th percentile error of $0.0624\,\mathrm{m}$ in Scene 1. Despite avoiding iterative nonlinear optimization, the proposed method achieves performance comparable to the AOA-TDOA NLS approach while substantially outperforming both the TDOA-only and AOA-only baselines, improving the MAE by 61.7% and 70.0%, respectively.

In terms of orientation estimation, Fig. 10(b) shows that SCALA achieves a MAE of $0.67°$ and a median error of $0.27°$. Although the introduction of TDOA constraints slightly affects orientation accuracy due to residual noise in certain TDOA measurements, the degradation remains marginal. Overall, the complementary fusion of AOA and TDOA constraints provides a more accurate, stable, and reliable solution for practical acoustic pose estimation.

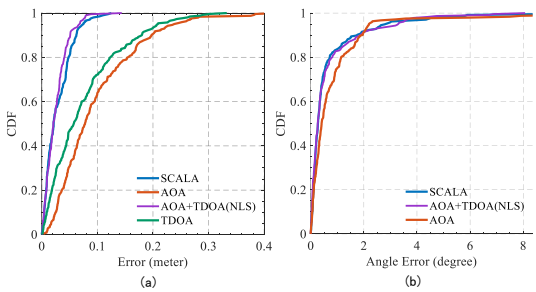


Fig. 10. Cumulative error distribution of position (left) and orientation estimates (right) for SCALA and its baselines in Scene 1.

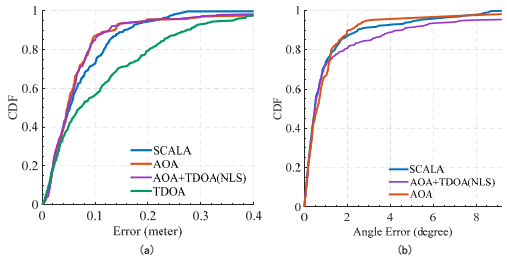


Fig. 11. Cumulative error distribution of position (left) and orientation estimates (right) for SCALA and its baselines in Scene 2.

D. Impact of Environment

To further evaluate the robustness of SCALA under acoustically challenging conditions, we introduce an additional experimental environment (Scene 2) with stronger multipath propagation and reverberation than Scene 1, as illustrated in Fig. 7d. This scenario contains more severe signal reflections and environmental clutter, providing a more realistic evaluation of practical indoor localization performance.

The experimental results in Scene 2 are summarized in Fig. 11. Compared with the relatively open environment in Scene 1, all methods exhibit degraded localization and orientation estimation performance due to stronger multipath interference and TOA distortion. This degradation is mainly reflected by an increased number of outlier frames, resulting in larger MAE and RMSE values while the median errors remain relatively stable. However, SCALA maintains relatively stable performance under the increased environmental complexity, particularly in orientation estimation, demonstrating stronger robustness against severe reverberation and multipath interference.

Despite the more challenging acoustic conditions, SCALA still maintains strong localization accuracy and robustness. In Scene 2, SCALA achieves a localization MAE of $0.0763\,\mathrm{m}$, an RMSE of $0.1329\,\mathrm{m}$, a median error of $0.0525\,\mathrm{m}$, and a 90th percentile error of $0.1574\,\mathrm{m}$. Compared with the TDOA baseline, SCALA reduces the localization MAE and RMSE by approximately 44.8% and 27.4%, respectively. Although the iterative AOA-TDOA (NLS) method achieves slightly lower localization error in this scenario, SCALA still maintains comparable performance while significantly reducing runtime and exhibiting more stable overall behavior under environmental degradation.

For orientation estimation, SCALA achieves a MAE of $1.12°$, an RMSE of $2.09°$, and a median error of $0.48°$, demonstrating comparable or better performance than the baseline methods. In particular, compared with the AOA-TDOA (NLS) method, SCALA achieves lower RMSE and improved robustness, indicating that the proposed closed-form hybrid formulation remains stable under severe reverberation and multipath conditions. These results further demonstrate that SCALA effectively combines the complementary advantages of AOA and TDOA constraints to achieve robust acoustic pose estimation.

TABLE III
PER-FRAME RUNTIME COMPARISON OF SIGNAL PROCESSING MODULES AND LOCALIZATION METHODS.

Signal Processing Modules	
Module	**Runtime (s)**
FBTE	8.1×10^{-4}
GCC (4-band)	1.99×10^{-3}
TABF	1.05×10^{-2}
Raw FrFT (4-band)	9.946×10^{-2}
AOA Constraint (Single Band)	8.0×10^{-4}
Localization Methods	
Method	**Runtime (s)**
TDOA	1.56×10^{-2}
AOA	1.45×10^{-2}
AOA-TDOA (NLS)	2.63×10^{-2}
SCALA (Proposed)	1.46×10^{-2}

E. Runtime Comparison

Efficient Batch TOA Extraction. Table III compares the per-frame runtime of the signal processing modules and localization algorithms. For TOA extraction, the proposed FBTE module requires only $8.1 \times 10^{-4}\,\mathrm{s}$ per frame, while the four-band GCC pipeline requires $1.99 \times 10^{-3}\,\mathrm{s}$, corresponding to a 59.3% runtime reduction. This improvement comes from the batch FMCW formulation, which jointly processes multi-band signals in the dechirped frequency domain and avoids repeated correlation operations across bands.

Reduced Multi-Channel Processing Overhead. A similar trend is observed in the AOA processing stage. The proposed TABF module requires $1.05 \times 10^{-2}\,\mathrm{s}$ per frame, whereas the Raw FrFT (4-band) pipeline requires $9.946 \times 10^{-2}\,\mathrm{s}$, achieving nearly one order of magnitude lower runtime. This gain mainly benefits from the proposed time-aligned batch processing strategy, which avoids repeated independent FrFT operations across channels and sub-bands.

Efficient Closed-Form Hybrid Localization. At the system level, SCALA achieves a per-frame runtime of $1.46 \times 10^{-2}\,\mathrm{s}$, which is comparable to the AOA-only and TDOA-only baselines while providing substantially better localization accuracy and robustness. Compared with the iterative AOA-TDOA (NLS) method, SCALA reduces the runtime by 44.5% while maintaining comparable localization performance. This improvement mainly comes from the proposed closed-form hybrid formulation, which avoids iterative nonlinear optimization. Overall, SCALA achieves an effective balance between localization accuracy and computational efficiency for low-latency acoustic localization.

VII. CONCLUSION

In this paper, we present SCALA, a temporal Self-Calibrated Asynchronous Localization Algorithm that jointly exploits AOA and TDOA observations for low-latency indoor localization. To the best of our knowledge, SCALA is the first passive acoustic localization system that unifies asynchronous AOA-TDOA fusion, online inter-anchor offset calibration, and closed-form pose estimation within a single framework. We develop FBTE and TABF for efficient TOA and AOA extraction from multi-anchor acoustic signals, and

further propose an online self-calibration strategy together with a closed-form hybrid AOA-TDOA formulation for joint pose estimation. Experimental results in real-world indoor environments demonstrate that SCALA achieves a median localization error of $0.0215\,\mathrm{m}$, a median orientation error of $0.27°$, and a per-frame runtime of only $1.46 \times 10^{-2}\,\mathrm{s}$, demonstrating an effective balance between accuracy, robustness, and computational efficiency.

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REFERENCES

[1] A. Qaddos, M. U. Yaseen, A. S. Al-Shamayleh, M. Imran, A. Akhunzada, and S. Z. Alharthi, "A novel intrusion detection framework for optimizing iot security," *Scientific Reports*, vol. 14, no. 1, p. 21789, 2024.

[2] Y. Sun, W. Wang, L. Mottola, J. Zhang, R. Wang, and Y. He, "Indoor drone localization and tracking based on acoustic inertial measurement," *IEEE Transactions on Mobile Computing*, vol. 23, no. 6, pp. 7537–7551, 2023.

[3] G. Liu, W. Cui, N. Jia, Y. Xi, S. Li, and Z. Wang, "Alcdnet: Loop closure detection based on acoustic echoes," *IEEE Robotics and Automation Letters*, 2024.

[4] A. Wang and S. Gollakota, "Millisonic: Pushing the limits of acoustic motion tracking," in *Proceedings of the 2019 CHI conference on human factors in computing systems*, pp. 1–11, 2019.

[5] J. Aparicio, F. J. Alvarez, A. Hernandez, and S. Holm, "A survey on acoustic positioning systems for location-based services," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–36, 2022.

[6] S. Yang, D. Zhang, R. Song, P. Yin, and Y. Chen, "Multiple wifi access points co-localization through joint aoa estimation," *IEEE Transactions on Mobile Computing*, vol. 23, no. 2, pp. 1488–1502, 2023.

[7] D. Yan, P. Yang, F. Shang, N. M. Freris, and Y. Yan, "Non-intrusive and efficient estimation of antenna 3-d orientation for wifi aps," *IEEE Transactions on Mobile Computing*, 2024.

[8] C. Zhang, X. Han, H. Kong, and K. Ho, "Optimal sensor placement for full-set tdoa localization accounting for sensor location errors," *IEEE Transactions on Aerospace and Electronic Systems*, 2025.

[9] A. Liu and W. Wang, "An indoor positioning method based on smartphone and surveillance camera," *IEEE Sensors Journal*, vol. 23, no. 19, pp. 22742–22753, 2023.

[10] Y. You and C. Wu, "Hybrid indoor positioning system for pedestrians with swinging arms based on smartphone imu and rssi of ble," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–15, 2021.

[11] F. Ge and Y. Shen, "Single-anchor ultra-wideband localization system using wrapped pdoa," *IEEE Transactions on Mobile Computing*, vol. 21, no. 12, pp. 4609–4623, 2021.

[12] Q. Zhang, K. Cui, C. Wang, and F. Xiao, "Tde-loc: Unambiguous rfid localization using trajectory deformation error," *IEEE Transactions on Instrumentation and Measurement*, 2025.

[13] X. Yang, Y. Zhuang, M. Shi, Q. Meng, J. Xiong, and Y. Cao, "Deepvlp: a graph neural network-based denoising and signals optimization framework for visible light positioning," *IEEE Transactions on Mobile Computing*, vol. 24, no. 6, pp. 5106–5123, 2025.

[14] W. Luo, Q. Song, Z. Yan, R. Tan, and G. Lin, "Indoor smartphone slam with acoustic echoes," *IEEE Transactions on Mobile Computing*, vol. 23, no. 6, pp. 6634–6649, 2023.

[15] L. Fu, Y. He, J. Wang, X. Qiao, and H. Kong, "I-asm: Iterative acoustic scene mapping for enhanced robot auditory perception in complex indoor environments," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 12318–12323, IEEE, 2024.

[16] K. Liu, X. Liu, and X. Li, "Guoguo: Enabling fine-grained indoor localization via smartphone," in *Proceeding of the 11th annual international conference on Mobile systems, applications, and services*, pp. 235–248, 2013.

[17] R. Chen, Z. Li, F. Ye, G. Guo, S. Xu, L. Qian, Z. Liu, and L. Huang, "Precise indoor positioning based on acoustic ranging in smartphone," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–12, 2021.

[18] C. Cai, R. Zheng, J. Li, L. Zhu, H. Pu, and M. Hu, "Asynchronous acoustic localization and tracking for mobile targets," *IEEE Internet of Things Journal*, vol. 7, no. 2, pp. 830–845, 2019.

[19] Y.-T. Wang, R. Zheng, and D. Zhao, "Towards zero-configuration indoor localization using asynchronous acoustic beacons," in *2016 IEEE Intl Conference on Computational Science and Engineering (CSE) and IEEE Intl Conference on Embedded and Ubiquitous Computing (EUC) and 15th Intl Symposium on Distributed Computing and Applications for Business Engineering (DCABES)*, pp. 32–39, IEEE, 2016.

[20] A. Gabbrielli, J. Bordoy, W. Xiong, G. K. Fischer, T. Schaechtle, J. Wendeberg, F. Höflinger, C. Schindelhauer, and S. J. Rupitsch, "Rails: 3-d real-time angle of arrival ultrasonic indoor localization system," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–15, 2022.

[21] S. Wang, X. Jiang, and H. Wymeersch, "Cooperative localization in wireless sensor networks with aoa measurements," *IEEE Transactions on Wireless Communications*, vol. 21, no. 8, pp. 6760–6773, 2022.

[22] G. K. Fischer, T. Schaechtle, A. Gabbrielli, J. Bordoy, I. Häring, F. Höflinger, and S. J. Rupitsch, "A systematic survey and comparative analysis of angular-based indoor localization and positioning technologies," *IEEE Communications Surveys & Tutorials*, 2025.

[23] H.-J. Shao, X.-P. Zhang, and Z. Wang, "Efficient closed-form algorithms for aoa based self-localization of sensor nodes using auxiliary variables," *IEEE Transactions on Signal Processing*, vol. 62, no. 10, pp. 2580–2594, 2014.

[24] M. Aernouts, N. BniLam, R. Berkvens, and M. Weyn, "Tdaoa: A combination of tdoa and aoa localization with lorawan," *Internet of Things*, vol. 11, p. 100236, 2020.

[25] C. Xu, Z. Wang, Y. Wang, Z. Wang, and L. Yu, "Three passive tdoa-aoa receivers-based flying-uav positioning in extreme environments," *IEEE Sensors Journal*, vol. 20, no. 16, pp. 9589–9595, 2020.

[26] D. Pang, G. Wang, and K. Ho, "Hybrid aoa-tdoa localization of a moving source by single receiver," *IEEE Transactions on Communications*, vol. 73, no. 6, pp. 4088–4104, 2024.

[27] X. Kang, D. Wang, Y. Shao, M. Ma, and T. Zhang, "An efficient hybrid multi-station tdoa and single-station aoa localization method," *IEEE Transactions on Wireless Communications*, vol. 22, no. 8, pp. 5657–5670, 2023.

[28] Y. Wang and K. Ho, "Unified near-field and far-field localization for aoa and hybrid aoa-tdoa positionings," *IEEE Transactions on Wireless Communications*, vol. 17, no. 2, pp. 1242–1254, 2017.

[29] J. Yin, Q. Wan, S. Yang, and K. Ho, "A simple and accurate tdoa-aoa localization method using two stations," *IEEE Signal Processing Letters*, vol. 23, no. 1, pp. 144–148, 2015.

[30] C. Zhang, W. Pan, X. Han, and H. Kong, "Calibration of multiple asynchronous microphone arrays using hybrid tdoa," in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, 2025.

[31] A. Ens, F. Höflinger, J. Wendeberg, J. Hoppe, R. Zhang, A. Bannoura, L. M. Reindl, and C. Schindelhauer, "Acoustic self-calibrating system for indoor smart phone tracking," *International journal of navigation and observation*, vol. 2015, no. 1, p. 694695, 2015.

[32] Y.-T. Wang, J. Li, R. Zheng, and D. Zhao, "Arabis: An asynchronous acoustic indoor positioning system for mobile devices," in *2017 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, pp. 1–8, IEEE, 2017.

[33] G. Li, L. Zhang, F. Lin, M. Chen, and Z. Wang, "Sailoc: A novel acoustic single array system for indoor localization," in *2017 9th international conference on wireless communications and signal processing (WCSP)*, pp. 1–6, IEEE, 2017.

[34] Z. Gao, A. Li, D. Li, J. Liu, J. Xiong, Y. Wang, B. Li, and Y. Chen, "Mom: Microphone based 3d orientation measurement," in *2022 21st ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*, pp. 132–144, IEEE, 2022.

[35] S. Ogiso, K. Mizutani, N. Wakatsuki, and T. Ebihara, "Robust indoor localization in a reverberant environment using microphone pairs and asynchronous acoustic beacons," *IEEE Access*, vol. 7, pp. 123116–123127, 2019.

[36] S. Cao, Y. Liu, S. Liu, Y. Liu, and P. Li, "Acoustic indoor localization using two base stations based on tdoa and relative speed to base stations," *IEEE Internet of Things Journal*, 2025.

[37] D. Hu and X. Wang, "Robust self-localization of wireless acoustic sensor networks in the presence of tdoa outliers," *IEEE Transactions on Mobile Computing*, 2026.
[38] P. Lazik, N. Rajagopal, O. Shih, B. Sinopoli, and A. Rowe, "Alps: A bluetooth and ultrasound platform for mapping and localization," in *Proceedings of the 13th ACM conference on embedded networked sensor systems*, pp. 73–84, 2015.
[39] S. Cao, X. Chen, X. Zhang, and X. Chen, "Effective audio signal arrival time detection algorithm for realization of robust acoustic indoor positioning," *IEEE Transactions on Instrumentation and Measurement*, vol. 69, no. 10, pp. 7341–7352, 2020.
[40] H. Ye, B. Yang, Z. Long, and C. Dai, "A method of indoor positioning by signal fitting and pdda algorithm using ble aoa device," *IEEE Sensors Journal*, vol. 22, no. 8, pp. 7877–7887, 2022.
[41] K. Hong, T. Wang, J. Liu, Y. Wang, and Y. Shen, "A learning-based aoa estimation method for device-free localization," *IEEE Communications Letters*, vol. 26, no. 6, pp. 1264–1267, 2022.
[42] J. Xiong and K. Jamieson, "{ArrayTrack}: A {Fine-Grained} indoor location system," in *10th USENIX Symposium on Networked Systems Design and Implementation (NSDI 13)*, pp. 71–84, 2013.
[43] F. Jacob, J. Schmalenstroeer, and R. Haeb-Umbach, "Doa-based microphone array postion self-calibration using circular statistics," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 116–120, IEEE, 2013.
[44] D. Zhang, J. Chen, J. Bai, and H. Yin, "Doa-based multi-node geometry calibration for wireless acoustic sensor network," in *2023 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC)*, pp. 1–5, IEEE, 2023.
[45] W. Mao, J. He, and L. Qiu, "Cat: High-precision acoustic motion tracking," in *Proceedings of the 22nd Annual International Conference on Mobile Computing and Networking*, pp. 69–81, 2016.
[46] M. Brandstein and D. Ward, *Microphone arrays: signal processing techniques and applications*. Springer Science & Business Media, 2001.
[47] J. Zhang, S. Singh, *et al.*, "Loam: Lidar odometry and mapping in real-time.," in *Robotics: Science and systems*, vol. 2, pp. 1–9, Berkeley, CA, 2014.

APPENDIX A
DERIVATION OF CONSTRAINTS

A. Derivation of the Linearized AOA Constraints

Starting from the AOA measurement model

$$\tan(\varphi+\theta_i)=\frac{b_i-y}{a_i-x}, \tag{55}$$

we rewrite it as

$$(a_i-x)\tan(\varphi+\theta_i)=b_i-y. \tag{56}$$

Using the angle addition identity

$$\tan(\varphi+\theta_i)=\frac{\tan\varphi+\tan\theta_i}{1-\tan\varphi\tan\theta_i}, \tag{57}$$

and after algebraic rearrangement, the measurement constraint becomes

$$\begin{aligned}&\sin\theta_i(x+y\tan\varphi)-\cos\theta_i(y-x\tan\varphi)\\&\quad-(a_i\cos\theta_i+b_i\sin\theta_i)\tan\varphi\\&=a_i\sin\theta_i-b_i\cos\theta_i.\end{aligned} \tag{58}$$

To linearize the nonlinear trigonometric coupling, define

$$\begin{aligned}u_1&=x\sin\varphi-y\cos\varphi,\\u_2&=x\cos\varphi+y\sin\varphi,\\u_3&=\sin\varphi,\\u_4&=\cos\varphi.\end{aligned} \tag{59}$$

Multiplying the resulting equation by u_4, we obtain the homogeneous linear AOA constraint

$$\begin{aligned}&\cos\theta_i u_1+\sin\theta_i u_2-(a_i\cos\theta_i+b_i\sin\theta_i)u_3\\&\qquad-(a_i\sin\theta_i-b_i\cos\theta_i)u_4=0.\end{aligned} \tag{60}$$

Stacking these constraints yields

$$\mathbf{M}_{\mathrm{AOA}}\mathbf{u}=\mathbf{0}.$$

B. Derivation of the Projected TDOA Constraints

Starting from the range-difference model

$$2\Delta a_i x+2\Delta b_i y+2d_{i1}r_1=H_i, \tag{61}$$

where

$$H_i=a_i^2+b_i^2-a_1^2-b_1^2-d_{i1}^2, \tag{62}$$

we project the constraint onto $\sin\varphi$ and $\cos\varphi$.

Multiplication by $\sin\varphi$ gives

$$2\Delta a_i(x\sin\varphi)+2\Delta b_i(y\sin\varphi)+2d_{i1}(r_1\sin\varphi)-H_i\sin\varphi=0, \tag{63}$$

while multiplication by $\cos\varphi$ yields

$$2\Delta a_i(x\cos\varphi)+2\Delta b_i(y\cos\varphi)+2d_{i1}(r_1\cos\varphi)-H_i\cos\varphi=0. \tag{64}$$

Using the lifted variable definition, the corresponding coefficient rows become

$$\mathbf{m}_{i,\mathrm{TDOA,sin}}^{\mathsf{T}}=\big[2\Delta a_i,\ 0,\ 2\Delta b_i,\ 0,\ 2d_{i1},\ 0,\ -H_i,\ 0\big], \tag{65}$$

and

$$\mathbf{m}_{i,\mathrm{TDOA,cos}}^{\mathsf{T}}=\big[0,\ 2\Delta a_i,\ 0,\ 2\Delta b_i,\ 0,\ 2d_{i1},\ 0,\ -H_i\big]. \tag{66}$$

Stacking the sine and cosine rows in alternating order forms the projected TDOA block $\mathbf{M}_{\mathrm{TDOA}}\mathbf{u}=\mathbf{0}$.

C. Sensitivity to Fusion Weight

Fig. 12 shows that the localization performance of SCALA remains stable under different η settings. As η varies from 0.4 to 1.2, the MAE fluctuation is limited to approximately 0.0022 m, while the RMSE and 90% error exhibit similarly consistent trends. The best performance is achieved at $\eta=0.8$ with an MAE of 0.0293 m. These results indicate that η mainly serves as a coarse balancing factor between AOA and TDOA constraints rather than a highly sensitive empirical parameter.

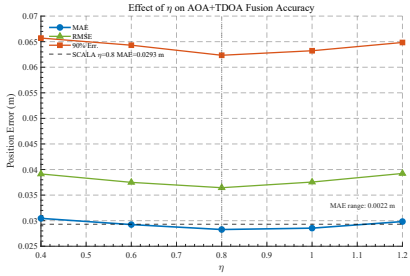


Fig. 12. Sensitivity analysis of the fusion parameter η in the proposed AOA+TDOA hybrid localization framework.